

UTHM Smart Fun Run: Centralized Dashboard Project

"United by Data, Driven by Innovation."

The Problem: Fragmented Data Silos



The Challenge of Isolated Data

Current methods rely on isolated silos, leading to communication gaps and a lack of real-time visibility during large-scale coordination.



Administrative Complexity

Tracking diverse student groups and academic schedules across university-wide activities is highly inefficient without a centralized platform.

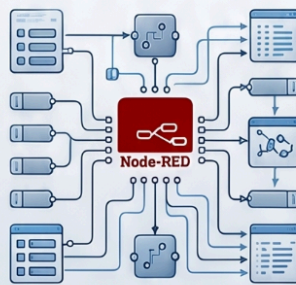
The 3-Tier IoT Architecture

Tier 1: Communication (Network Infrastructure)



The system relies on high-speed Wi-Fi infrastructure to facilitate the real-time transmission and consolidation of diverse data streams.

Tier 2: Middleware (Node-RED Integration)



Node-RED acts as the central engine to process logic and bridge the gap between isolated data silos, creating a cohesive operating system.

Tier 3: Application (Centralized Dashboard)



A streamlined, user-friendly monitoring platform provides the final interface for event organizers to view consolidated data.

Key Features & Outcomes



Real-Time Participant Tracking

Organizers gain immediate visibility into runner progress and registration status, replacing manual, fragmented tracking methods.



Enhanced Logistics Management

The dashboard serves as the primary tool for managing event operations, ensuring efficient resource allocation and data collection.



Consolidated Data Streams

Multiple subject areas and student group data are merged into one view to eliminate miscommunication.

Project Identity

Developed by Group A3



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